

Mark Higham

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Contract full-stack developer with 20+ years' experience delivering software for defence, central government and global enterprises. Currently building React and Python applications and the Royal Navy Design System in secure, air-gapped environments. Equally effective hands-on or as a technical lead, working solo, in a pair or in larger agile teams.

Twitter: @markhigham

LinkedIn: <https://www.linkedin.com/in/markhigham/>

CV: <https://github.com/markhigham/cv>

Contract experience in brief

Contract Developer – Digi2al Ltd – Jan 2023 – Present

Full-stack developer. TypeScript: React/Next.js, Storybook; Python: Flask, Django, DRF, Wagtail; Royal Navy Design System; Palantir Foundry, OSDK, PWA; WinTAK, C#

Contract Developer – Vaultex UK – Aug 2022 – Jan 2023

C#, .NET Core Azure Function apps, SQL Server, SpecFlow

Data Engineer – Department for International Trade – May 2021 – Aug 2022

Python, Django, Django REST framework, Postgres, Apache Airflow, Docker, AWS

Contract Developer – Department for International Trade – May 2019 – May 2021

Python, Django, Flask, Postgres, Docker, Redis, AG Grid, Elasticsearch, SharePoint, Azure DevOps, Cloud Foundry (GOV.UK PaaS)

Contract Technical Lead – DXC Enterprise Security Services – Mar 2017 – Mar 2019

Angular, React, Sass, Python, C#, Java, ASP.NET, NancyFx, SQL Server, AG Grid, Jenkins, MSBuild, Docker, Vagrant, ServiceNow

Technical Lead – Ruler Analytics – Jan 2016 – Mar 2017

Python, Django, Node.js, TypeScript, Elasticsearch, AG Grid, MySQL, Redis, RabbitMQ, Angular, AngularJS, TeamCity, Ansible, Docker, Vagrant, AWS EC2, RDS, ElastiCache, SQS, SNS, Lambda, API Gateway, Serverless

Full Stack Developer – HPE – May 2015 – Dec 2015

ASP.NET, C#, Node.js, MongoDB, Redis, RabbitMQ, SQL Server, Docker, Vagrant, Jenkins

Contract Developer – Swinton Insurance – May 2014 – May 2015

ASP.NET, WebAPI, WCF, NServiceBus, SQL Server, RabbitMQ, MSMQ, TeamCity CI/CD

Solutions Architect – slicedbread – Oct 2013 – Apr 2014

ASP.NET, Orchard CMS, SQL Server, Azure, user research, client visits

Contract Developer – NICE – Feb 2013 – Jun 2013

ASP.NET MVC 4, Orchard CMS, SQL Server, TeamCity

Development Lead – Pulse Umbrella – Aug 2009 – Oct 2012

ASP.NET, SQL Server, Node.js, Excel VBA, integration with Sage

Previous Management Roles

- 2007 – 2009 IT Operations Manager – Blemain Finance
- 2003 – 2007 Development Manager – Paymentsshield Ltd.
- 2001 – 2003 Project Lead – HBOS plc.

Education / Training

- PRINCE2 Practitioner – 2009
 - Certified Scrum Master – 2005
 - BSc Hons Economics – University of Wales Cardiff
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Contract Experience in Full

Contract Developer – Digi2al Ltd

January 2023 – Present

Building and supporting defence applications within a secure, air-gapped environment using React and Python.

- Built an offline-capable reporting progressive web app (PWA) on Palantir Foundry using the Ontology SDK (OSDK), deployed across the fleet
- Develop and support the [Royal Navy Design System](#), the shared design system used by multiple projects across the Royal Navy
- Rapidly prototyped a Furthest on Circle plugin for the WinTAK platform in C# as a proof of concept
- Ship React front-end and Python back-end features, defect fixes and CVE patches

Contract Developer – Vaultex UK Ltd

August 2022 – January 2023

Building cash-handling APIs using C# Azure Functions with SQL Server and Azure DevOps.

My role:

- Create work breakdown and support tech lead in creating Jira tickets
- Build and unit test the C# Azure Functions and SQL stored procedures
- Build Azure DevOps pipelines

Data Engineer – Department for International Trade (Data Infrastructure)

May 2021 – August 2022

Python, Django, Postgres, Apache Airflow, Docker, AWS Stack

The Data Infrastructure team build and support the data platforms in use by the [Department for International Trade](#) (now the Department for Business and Trade).

My role:

- Build Airflow pipelines – DAGs written in Python to ingest "big data" from third-party platforms including GCP, AWS S3, Postgres, UK ONS
- Work with UX / user researchers and designers to build new features
- Development of front-end features – Django / Python / JavaScript

Contract Developer – Department for International Trade (Live Services)

May 2019 – May 2021

Python, Django, Flask, Postgres, Docker, Redis, SharePoint, Azure DevOps

The Live Services team manage the ongoing support and maintenance of applications and platforms built within feature teams. For this reason, it requires technical staff with a wide experience of languages, platforms and tools (Python, JavaScript, Django, DRF, PostgreSQL, Redis, Elasticsearch, Docker, Jenkins, CircleCI, React, Vue, Webpack, Node.js, AG Grid, SharePoint).

Recent projects:

- UKDSE Events Planning – simple Python Django web application to support the planning of trade events.
- Insourcing Python/Django/Postgres/Elastic platform to support the management of Trade Remedies. <https://www.trade-remedies.service.gov.uk/>
- Insourcing of critical SharePoint EU Exit support platform from 3rd party supplier enabling significant cost savings.
- Supporting teams in defect fixing and deployment of critical Brexit platforms for fixed Brexit deadline <https://www.check-duties-customs-exporting-goods.service.gov.uk/selectdest>
- Insourcing and redevelopment of externally hosted knowledge base to a Wagtail / Django / Postgres application.

Contract Technical Lead – DXC Enterprise Security Services

March 2017 – March 2019

TypeScript / Node.js, Angular, React, Sass | C#, Java, ASP.NET, NancyFx, SQL Server, VB.NET | Jenkins, MSBuild, Docker, Vagrant | ServiceNow

Lead developer for multiple integrations with third-party security and ITSM platforms (ServiceNow, ArcSight SIEM, Qualys) using Python, C#, Java and Node.js running on Windows and Linux hosts.

Introduced TDD within the development team and built a "ticket exchange" between a VB.NET legacy web application and the ServiceNow REST API.

- Frontend stack: Angular served by ASP.NET, Browserify, Sass
- Backend: components built as microservices written in C# with Rebus / SQL Server for messaging and co-ordination

Created Jenkins pipelines and automated SQL Server updates using FluentMigrator.

Co-ordinated development and QA activity across US, UK, India and Australia with senior management teams across the globe and maintained Jira tasks and boards.

Technical Lead – Ruler Analytics

January 2016 – March 2017

Python, Django, Node.js, TypeScript | Elasticsearch, MySQL, Redis, RabbitMQ | Angular, AngularJS | TeamCity, Ansible, Docker, Vagrant | AWS EC2, RDS, ElastiCache, SQS, SNS, Lambda, API Gateway, Serverless

Ruler Analytics provide a real-time, big data, visitor analytics platform for their customers. We built it with Angular and it incorporates Chart.js data visualisations as well as sophisticated search and aggregation tools built upon AG Grid. Real-time data is provided by an Elasticsearch cluster via the Python Django REST framework.

Data analytics is built with Python Celery workers using a RabbitMQ broker and Redis results backend. Integrations with third parties are built with TypeScript / Node.js and Express, run as AWS Lambda functions and are orchestrated via SNS.

Visitor website tracking is managed by a lightweight native JavaScript tracking snippet.

I led the design, build and release of the Ruler Analytics platform: a distributed system composed of multiple microservices written in Python running via Celery over a RabbitMQ backbone, operating at thousands of events per second.

Built integrations with key platforms (Salesforce, Microsoft Dynamics, Intercom).

Designed and implemented CI pipelines using Git, TeamCity, Ansible and Serverless to the dev and production platforms.

Managed the migration of the entire platform to the AWS UK region.

Full Stack Developer – HPE

May 2015 – December 2015

Node.js, MongoDB, Redis, RabbitMQ | ASP.NET, C#, SQL Server | Docker, Vagrant | Jenkins

A key member of a globally-distributed technical team responsible for integrating legacy applications with new architecture. This role required a large amount of flexibility and a diverse skill set.

Designed and built a Node.js API interface to a legacy VB.NET / SQL Server / T-SQL application. This required collaboration with developers, analysts and product owners as well as reviewing, maintaining and updating legacy SQL Server code.

Contract Developer – Swinton Insurance

May 2014 – May 2015

ASP.NET, WebAPI, WCF, NServiceBus | SQL Server | SpecFlow, Cucumber, NUnit, JMeter

Key member of an agile development and DevOps team.

Responsible for the design and build of enterprise microservices for the management of insurance policy documents.

Automated the acceptance testing of PDF document production by building a PDF testing library.

Built a TeamCity dashboard via Node.js and the TeamCity API enabling the monitoring of over 2000 builds.

Integrated Excel with the TeamCity REST API (yes really!) allowing the PMs to view build versions of all enterprise services throughout the deployment pipelines.

Solutions Architect – slicedbread

October 2013 – April 2014

Orchard CMS, ASP.NET MVC 4, LESS | NUnit | Azure | Team Foundation Service

slicedbread.co.uk is a technology consultancy company based in Altrincham, Cheshire.

I was engaged to provide specialist development support for an Azure-hosted Orchard CMS site for a key UK plc client. The site was delivered within timescales (against the odds and the expectations of the client!)

Build of custom Orchard modules for re-use within slicedbread.

Contract Developer – NICE

February 2013 – June 2013

ASP.NET MVC 4, SQL Server, Orchard CMS, TeamCity, Bootstrap, LESS | SQL Server 2008 | IIS 7

National Institute for Health and Care Excellence (NICE) provide guidance to healthcare professionals. Their development team in Manchester is known for being early adopters of technologies and approaches. They build and ship their products in 2-weekly increments. Development is test-driven. Builds are automated.

Redevelopment of NICE intranet from ColdFusion to Orchard CMS. Development of custom Orchard modules (C#, MVC, Razor). Configuration of TeamCity and automation of builds.

Development Lead – Pulse Umbrella

August 2009 – October 2012

C#, ASP.NET Web Forms, SQL Server, JavaScript

Design, test and development of online customer portals and back-office applications.

Development and management of product backlog.

Design and UX testing of Pulse Umbrella's Contractor Portal.

Integration of Salesforce CRM platform with Pulse back-office systems using C# and WCF to provide a single view 'customer onboarding' process.

Integration of back-office systems using SQL Server and Sage Payroll and Accountancy packages for the straight-through processing of contractor pay and management of supplier invoices.